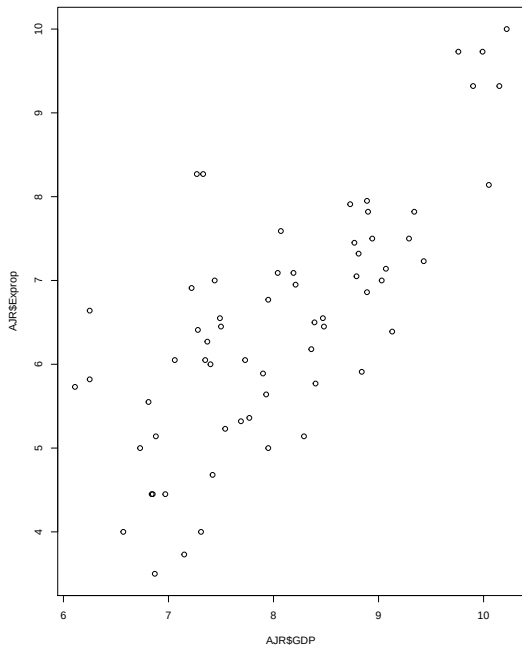


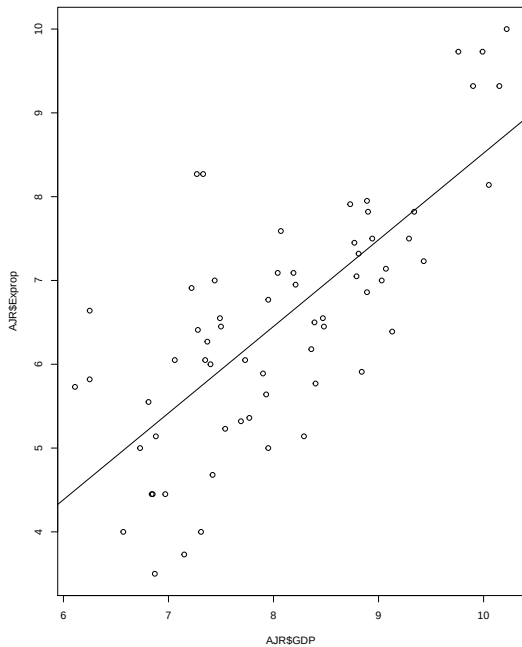
Linear Regression Analysis in the Social Sciences

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Session 2: Introduction to OLS





Regression

- Regression allows to examine relationships between several variables.
- Regression draws a line to best fit the data.
- Lines provide information about relationships: intercept and the slope.
- $y = \alpha + \beta x$
- Intercept (constant) α is the value of y when x equals zero. Slope β is the rise over the run. α and β are unknown parameters. We will try to estimate these using data on hand.

Regression

- Y_i : the dependent variable or outcome or regressand or left-hand-side variable or response
 - ▶ Expropriation Risk
 - ▶ Voter turnout
 - ▶ Number of battle deaths
- X_i : the independent variable or explanatory variable or regressor or right-hand-side variable or treatment or predictor
 - ▶ Log GDP per capita
 - ▶ Social pressure mailer versus Civic Duty Mailer
 - ▶ Presence of conflict mediation

What Does Regression Estimate?

- We observe pairs (Y_i, X_i) — an outcome and a predictor, measured on the same unit i .
- The question: **how does the average of Y change as X changes?**
 - ▶ E.g. A student asks you what grade they should expect in your course. That student has taken another one of your courses and received an A.
- This is the **conditional expectation function (CEF)**:

$$E[Y_i | X_i = x]$$

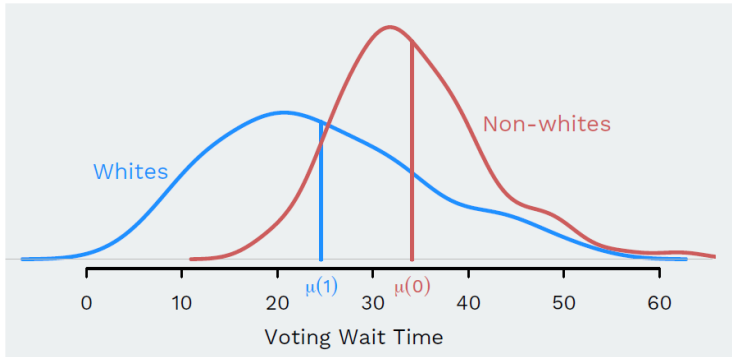
- Read this as: “the expected value of Y for units where X equals x .”, where x is a specific value of X (e.g. X = grade student received in your previous course, $x = \text{“A”}$)
- The goal of regression is to estimate this function.

CEF with a Binary X

- Y_i : time respondent i waited in line to vote.
- $X_i = 1$ for whites, $X_i = 0$ for non-whites.
- The CEF here is just two group means:

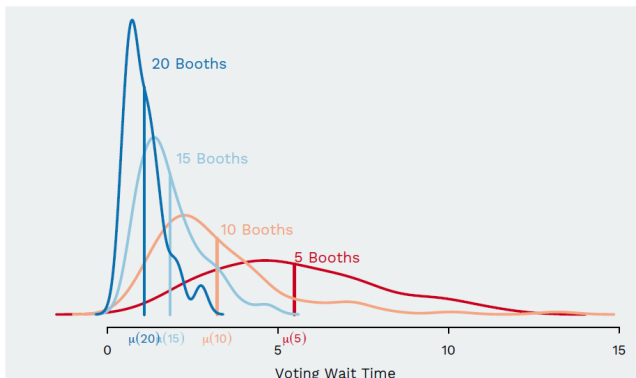
$$E[Y_i | X_i = \text{white}] \quad \text{and} \quad E[Y_i | X_i = \text{non-white}]$$

- Since X can only take two values, these two conditional means completely describe how wait times vary with race.
- If $E[Y_i | \text{white}] < E[Y_i | \text{non-white}]$, whites wait less on average.



- The CEF summarizes relationships between variables
- If $\mu(\text{white}) < \mu(\text{non} - \text{white})$, so that waiting times for whites are shorter on average than for non-whites.
- Indicates a relationship in the population (θ) between race and wait times.

CEF for discrete covariates



- Alternative covariate: X_i is the number of polling booths at citizen i 's polling station.
- The mean of Y_i changes as X_i changes

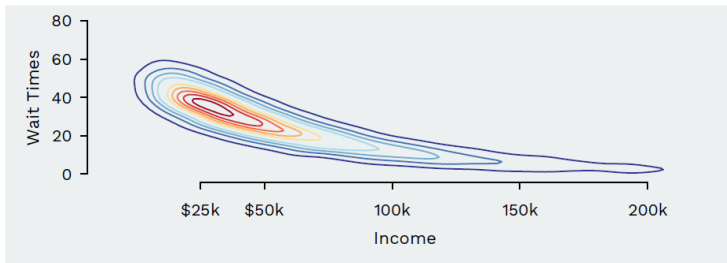
CEF with a Continuous Covariate

- What if X_i is income? Many possible values of X .
- We can't just list out each group mean — we need a *function* to summarize how the mean of Y changes with X .
- A natural choice: a linear function.

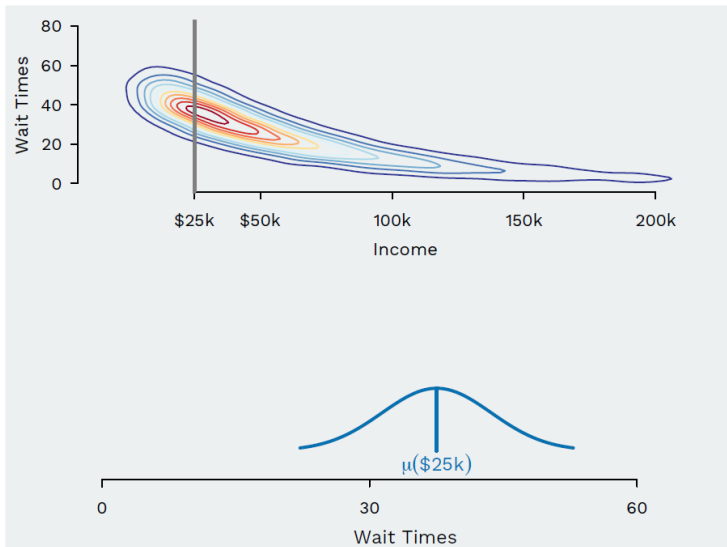
$$E[Y_i | X_i = x] \approx \alpha + \beta x$$

- α and β are unknown parameters. Our goal is to estimate them.

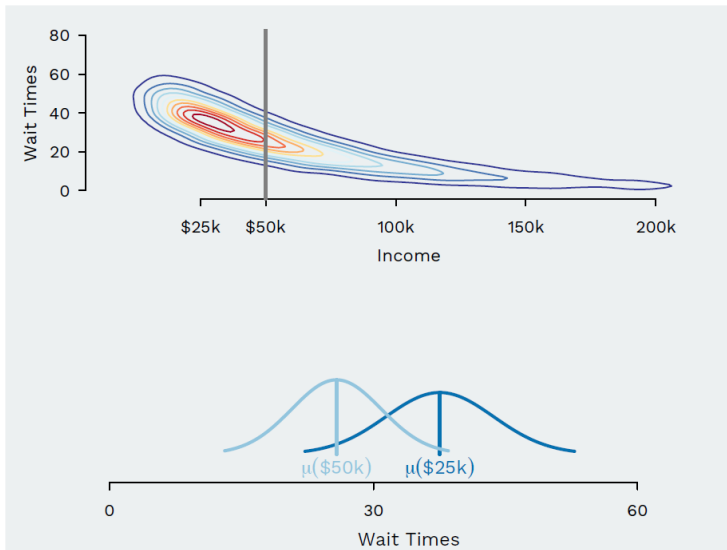
Wait times and income



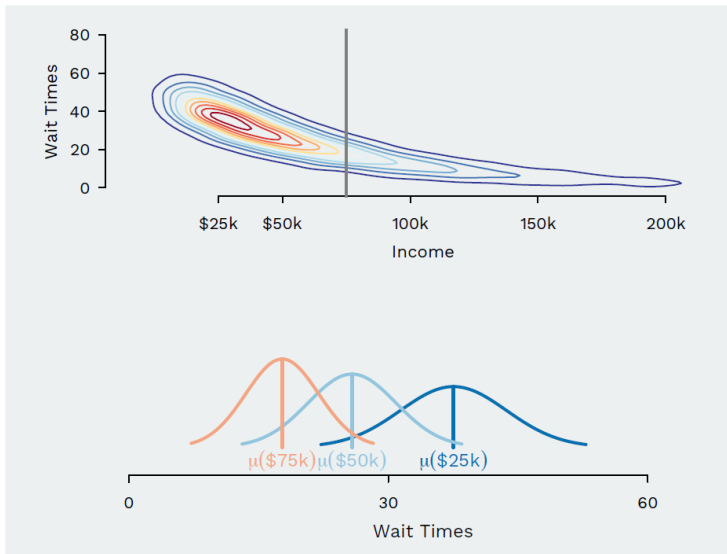
Wait times and income



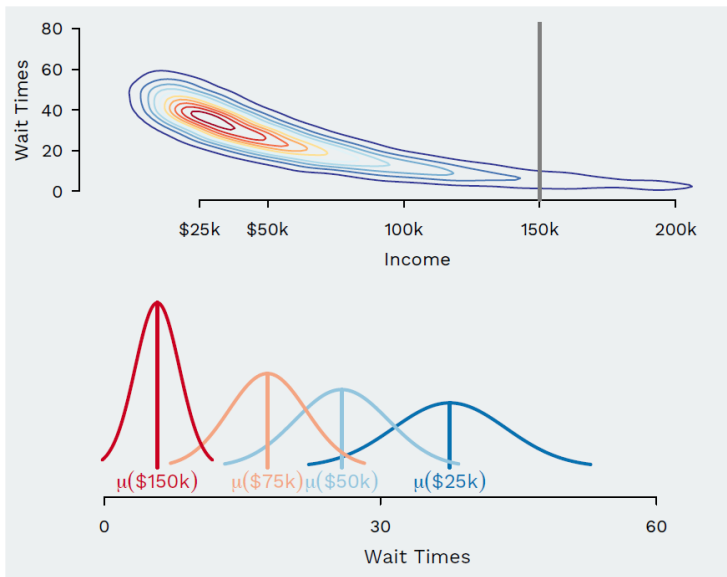
Wait times and income



Wait times and income



Wait times and income



Linear CEFs

- Intercept, α or β_0 : the condition expectation of Y_i when $X_i = 0$
 - ▶ Note, this isn't always of interest, for example when $X = 0$ is impossible or far outside of the range of data. (X is blood pressure, or height etc.)
 - ▶ Also note that adding something to your X values by some value, let's say a , changes the intercept, but not the slope.
 - ▶ Often a is set to $\bar{X}$ so that the intercept is interpreted as the expected response at the average X value.
- Slope, β_1 : average change in the mean of Y_i given a one-unit change in X_i .
 - ▶ Multiplication of X by a factor a results in dividing the coefficient by a factor of a .
 - ▶ Dividing X by its sd results in the coefficient representing a correlation coefficient.

Regression

- Example: X is height in m and Y is weight in kg . Then β_1 is kg/m . Converting X to cm implies multiplying X by 100 (given the ratio of cm/m). To get β_1 in the right units, we have to divide by $100cm/m$ to get it to have the right units.

$$Xm \times \frac{100cm}{m} = (100X)cm \quad \text{and} \quad \beta_1 \frac{kg}{m} \times \frac{1m}{100cm} = \left(\frac{\beta_1}{100} \right) \frac{kg}{cm}$$

Linearity as an assumption

- Example: Y_i is income, X_i is years of education
 - ▶ β_0 : average income among people with 0 years of education.
 - ▶ β_1 : expected difference in income between two adults that differ by 1 year of education.
- Linearity as an assumption:

$$\beta_1 = E[Y_i|X_i = 12] - E[Y_i|X_i = 11] = E[Y_i|X_i = 16] - E[Y_i|X_i = 15]$$

- Effect of getting HS degree is the same as the effect of getting college degree.

Fitting the best line example

- Let Y_i be the i^{th} child's height and X_i be the i^{th} (average over the pair of) parents' heights.
- Consider finding the best line
- Child's Height = β_0 + Parents' Heights β_1
- Use least squares to estimate our unknown parameters

$$\sum_{i=1}^n \{Y_i - (\beta_0 + \beta_1 X_i)\}^2$$

Fitting the best line

- The least squares model fits the line $Y = \hat{\beta}_0 + \hat{\beta}_1 X$ where

$$\hat{\beta}_1 = \text{Cor}(Y, X) \frac{Sd(Y)}{Sd(X)} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

- $\hat{\beta}_1$ has the units of Y/X .
- The line passes through the point $(\bar{X}, \bar{Y})$
- If you normalized the data, $\{\frac{Y_i - \bar{Y}}{Sd(Y)}, \frac{X_i - \bar{X}}{Sd(X)}\}$, the slope is $\text{Cor}(Y, X)$.

Covariance

To derive the OLS slope, we first need to understand how two variables move together.

The **covariance** measures whether X and Y tend to be jointly above (or below) their means:

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- $\text{Cov}(X, Y) > 0$: when X is above its mean, Y tends to be above its mean too.
- $\text{Cov}(X, Y) < 0$: they tend to move in opposite directions.
- $\text{Cov}(X, Y) = 0$: no linear association.

Covariance: Worked Example

Five countries, where X = years of education and Y = income (thousands):

Country	X_i	Y_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})(Y_i - \bar{Y})$
A	6	20	-4	-16	64
B	8	30	-2	-6	12
C	10	35	0	-1	0
D	12	40	2	4	8
E	14	55	4	19	76
$\bar{X} = 10$ $\bar{Y} = 36$			$\sum = 160$		

$$\text{Cov}(X, Y) = \frac{160}{5 - 1} = 40$$

Every product in the last column is positive or zero — when X is above its mean, Y is too.

From Covariance to Correlation

Covariance depends on scale — doubling X doubles the covariance.

Correlation removes this:

$$r_{X,Y} = \frac{\text{Cov}(X, Y)}{s_X \cdot s_Y}$$

Continuing our example:

- $\text{Cov}(X, Y) = 40$
- $s_X = \sqrt{\frac{(16+4+0+4+16)}{4}} = \sqrt{10} \approx 3.16$
- $s_Y = \sqrt{\frac{(256+36+1+16+361)}{4}} = \sqrt{167.5} \approx 12.94$
- $r_{X,Y} = \frac{40}{3.16 \times 12.94} \approx 0.98$

Strong positive correlation: education and income move closely together in these data.

Correlation: Properties

- Bounded: $-1 \leq r \leq 1$
- $r = 1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r = 0$: no linear association
- Unitless — doesn't depend on how we measure X or Y

Now notice: the OLS slope formula uses exactly these ingredients.

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{40}{10} = 4$$

A one-year increase in education is associated with a 4-thousand increase in income. We'll unpack this formula next.

The OLS Slope

The least squares solution gives us:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Look at this formula carefully:

- **Numerator:** how X and Y vary together — the covariance (up to scaling).
- **Denominator:** how much X varies — the variance of X (up to scaling).

$$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

The OLS slope is the covariance of X and Y , scaled by the variance of X .

Connecting Correlation and the Slope

We can also write:

$$\hat{\beta}_1 = \text{Cor}(Y, X) \times \frac{\text{Sd}(Y)}{\text{Sd}(X)}$$

Two components:

- $\text{Cor}(Y, X)$: the direction and strength of the linear relationship (scale-free, between -1 and 1).
- $\frac{\text{Sd}(Y)}{\text{Sd}(X)}$: converts from the scale-free correlation to the units of Y per unit of X .

If you standardize both variables (subtract mean, divide by SD), the slope *is* the correlation.

The intercept: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, so the line always passes through $(\bar{X}, \bar{Y})$.

Hands-On Example: Step by Step

Data: $(X_i, Y_i) = (1, 2), (2, 4), (3, 6)$

Step 1: Calculate means

- $\bar{X} = \frac{1+2+3}{3} = 2$
- $\bar{Y} = \frac{2+4+6}{3} = 4$

Step 2: Calculate deviations and cross-products

X_i	Y_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	2	-1	-2	2
2	4	0	0	0
3	6	1	2	2

Step 3: Apply formula

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{2 + 0 + 2}{1 + 0 + 1} = \frac{4}{2} = 2$$

Quick Check: Can You Calculate This?

Practice Problem:

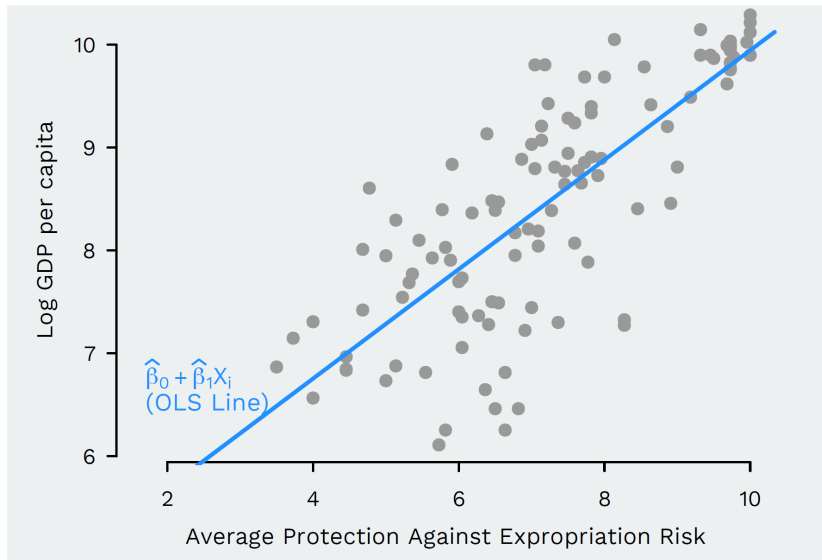
Given data: $(X, Y) = (0, 1), (1, 3), (2, 5)$

Calculate:

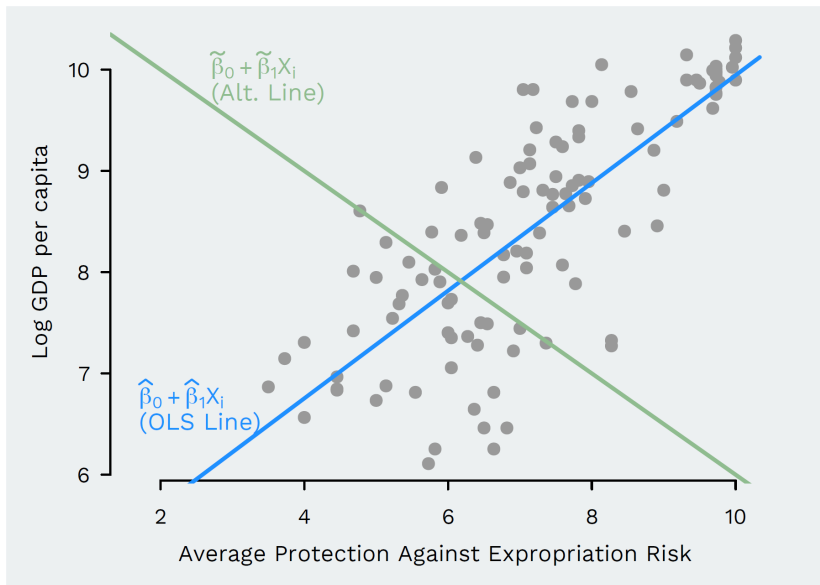
- 1 $\bar{X}$ and $\bar{Y}$
- 2 $\sum (X_i - \bar{X})(Y_i - \bar{Y})$
- 3 $\sum (X_i - \bar{X})^2$
- 4 $\hat{\beta}_1$

Answer: $\hat{\beta}_1 = 2$

Fitted Line Example



Fitted Line Example



Fitted values and residuals

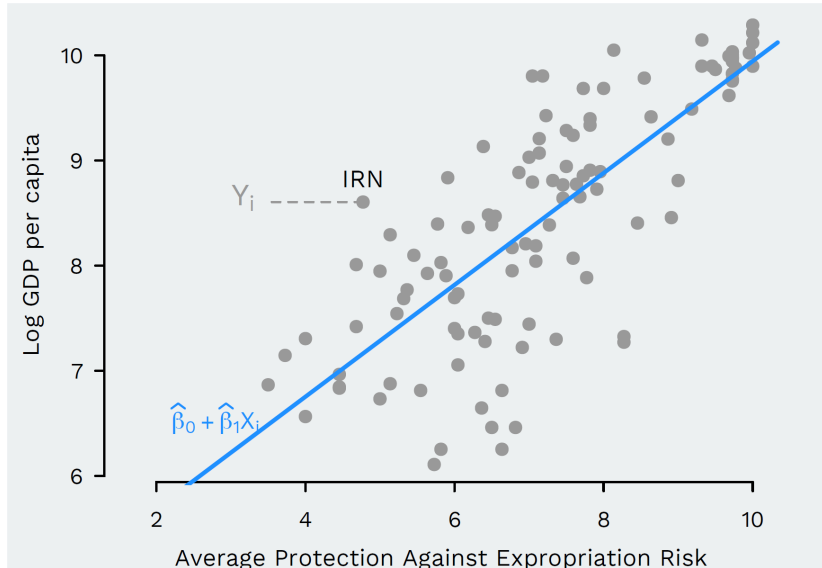
- Definition: A **fitted value** is the estimated conditional mean of Y_i for a particular observation with independent variable X .

$$\hat{Y}_i = \hat{E}[Y_i|X_i] = \hat{B}_0 + \hat{B}_1 X_1$$

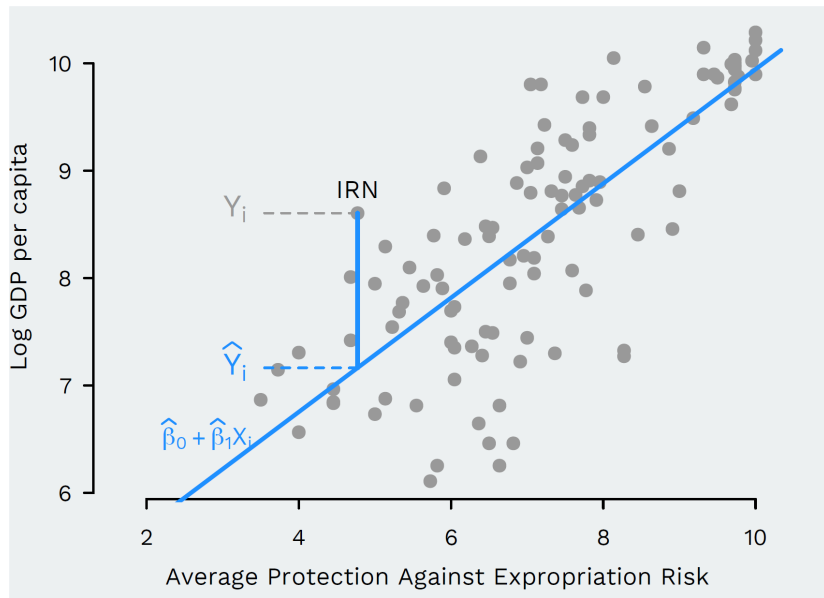
- Definition The residual is the difference between the actual value of Y_i and the fitted value, $\hat{Y}_i$:

$$\hat{e}_i = Y_i - \hat{Y}_i = Y_i - \hat{B}_0 - \hat{B}_1 X_1$$

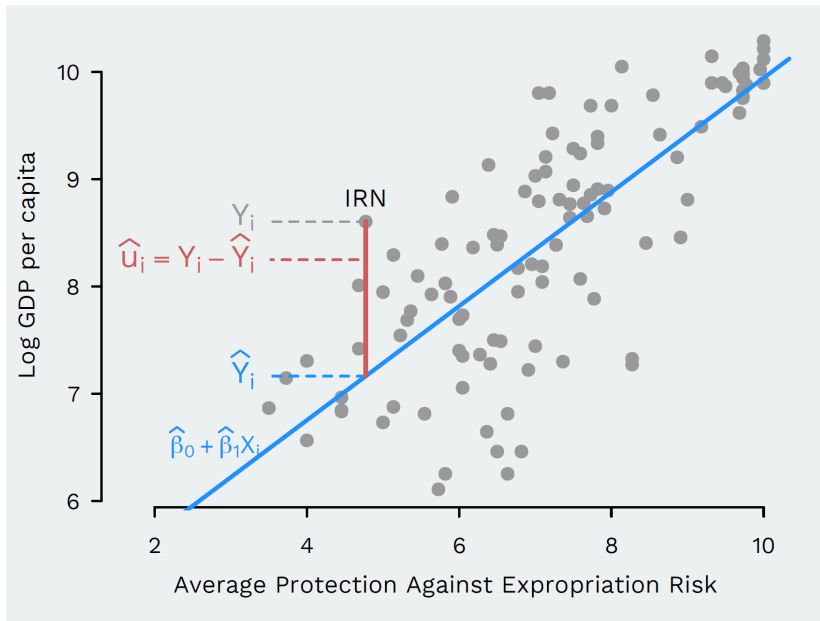
Fitted Line Example



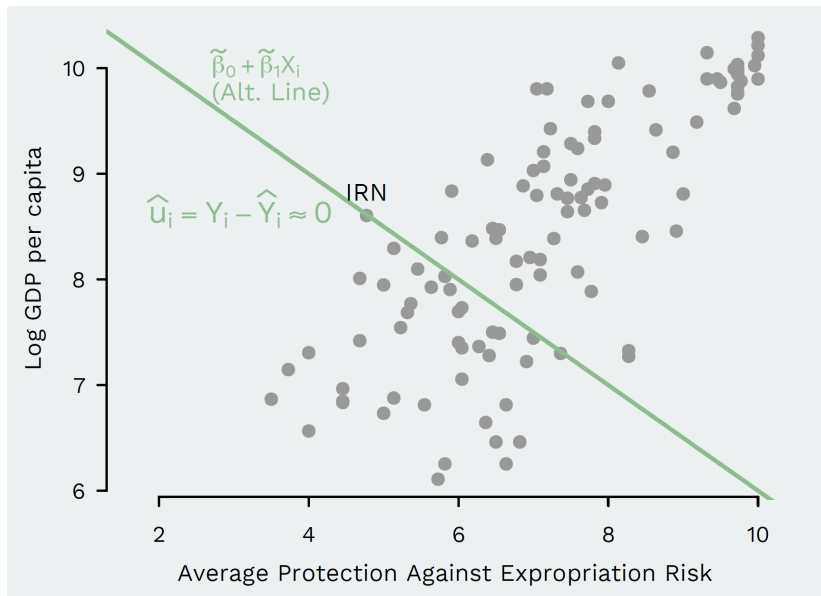
Fitted Line Example



Fitted Line Example



Fitted Line Example

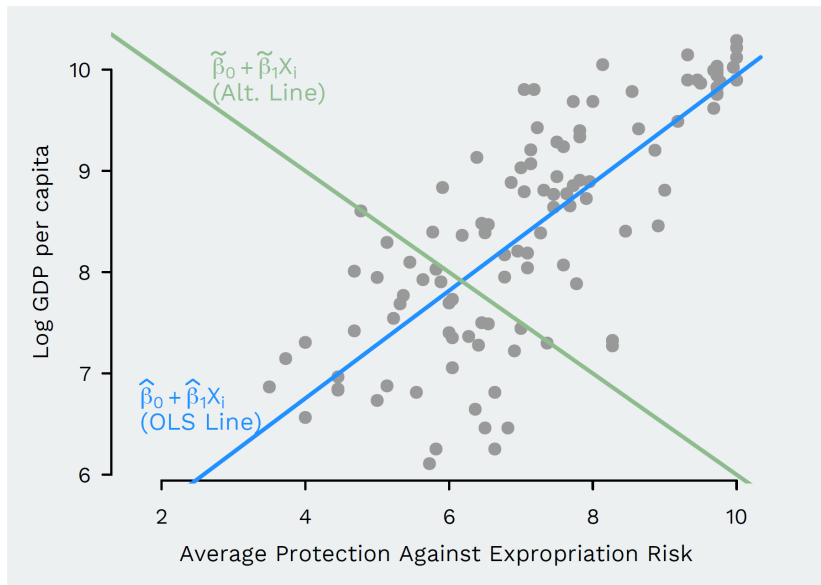


The residuals

The residuals, $\hat{e}_i$, tell us how well the line fits the data.

- Larger magnitude residuals means that points are very far from the line
- Residuals close to 0 mean points very close to the line
- The smaller the magnitude of the residuals, the better we are doing at predicting Y_i
- Choose the line that minimizes the residuals

Which line is best?



Regression Summary

- $\hat{\beta}_1$ has the units of Y/X .
- The line passes through the point $(\bar{X}, \bar{Y})$
- If you normalized the data, $\{\frac{Y_i - \bar{Y}}{Sd(Y)}, \frac{X_i - \bar{X}}{Sd(X)}\}$, the slope is $Cor(Y, X)$.

The CEF decomposition

- We can also decompose Y_i into the CEF and an error:

$$Y_i = E[Y_i|X_i] + \varepsilon$$

- Y_i can be decomposed into the part “explained by X_i ” and a part that is uncorrelated with X_i .
 - ▶ $E[\varepsilon] = 0$
 - ▶ $E[\varepsilon|X_i] = 0$
 - ▶ The error is uncorrelated with any function of X_i

Residuals

- Model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ where $\epsilon_i \sim N(0, \sigma^2)$.
- Observed outcome for i is Y_i at predictor value X_i
- Predicted outcome for i is $\hat{Y}_i$ at predictor value X_i is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

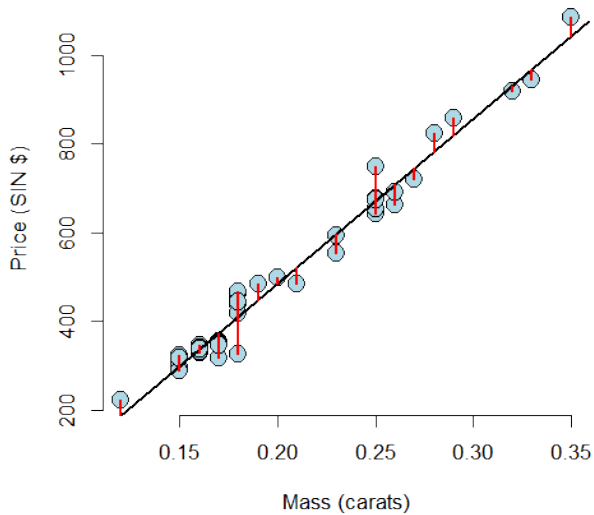
- Residual, the difference between the observed and predicted outcome

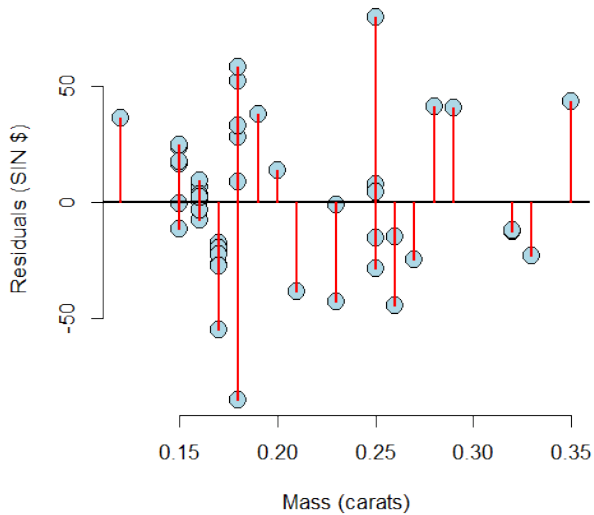
$$e_i = Y_i - \hat{Y}_i$$

- The vertical distance between the observed data point and the regression line
- Least squares minimizes $\sum_{i=1}^n e_i^2$
- The e_i can be thought of as estimates of the ϵ_i .

Properties of the residuals

- $E[e_i] = 0$.
- Residuals are useful for investigating poor model fit.
- Positive residuals are above the line, negative residuals are below.
- One differentiates residual variation (variation after removing the predictor) from systematic variation (variation explained by the regression model).
- Residual plots highlight poor model fit.





Estimating residual variation

- The total variability in our response is the variability around an intercept (think mean only regression) $\sum_{i=1}^n (Y_i - \bar{Y})^2$
- The regression variability is the variability that is explained by adding the predictor $\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$
- The error variability is what's leftover around the regression line $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$



$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

R^2 Statistic

- We have a statistic that tells us how well our model (our x variables) captures variation in the data, our dependent variable — R^2 .
- How much better are we able to predict y with x than without it (i.e. simply taking the mean of y as a prediction of y)?
- Proportional reduction in error.
- Be careful with this: it's useful for telling us how well we are predicting Y , but does not tell us about any specific relationship between X and Y .

R squared

- R squared is the percentage of the total variability that is explained by the linear relationship with the predictor

$$R^2 = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

R squared

- R^2 is the percentage of variation explained by the regression model.
- $0 \leq R^2 \leq 1$
- R^2 is the sample correlation squared.
- R^2 can be a misleading summary of model fit.
- Adding terms to a regression model always increases R^2 .

The relation between R squared and r

Recall that $(\hat{Y}_i - \bar{Y}) = \hat{\beta}_1(X_i - \bar{X})$ so that

$$R^2 = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = \hat{\beta}_1^2 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = \text{Cor}(Y, X)^2$$

Since, recall,

$$\hat{\beta}_1 = \text{Cor}(Y, X) \frac{\text{Sd}(Y)}{\text{Sd}(X)}$$

So, R^2 is literally r squared.